GEOMETRY · UNIT 9: CIRCLES

Pizza Parlor Geometry

Use circle area, arcs, sectors, and angles to price and slice pizza.

Honors

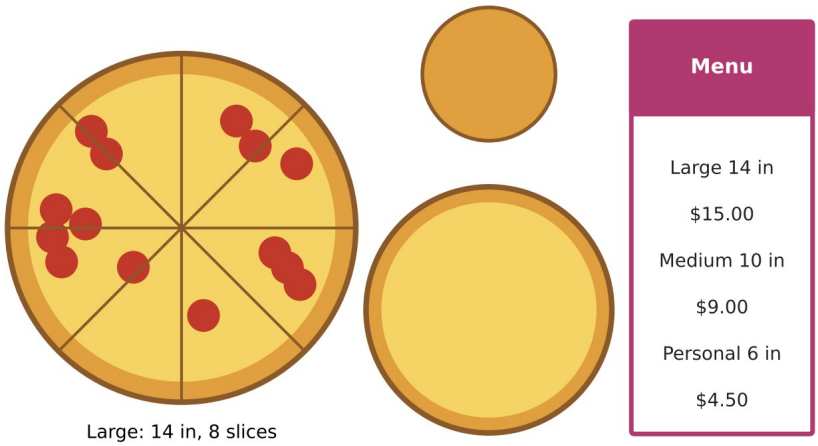
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Slice of Pi Pizzeria wants to update its menu, and customers keep asking: is two mediums really more pizza than one large?

Every question on the menu is a circle question: area, crust length, slice size, and delivery range. You will answer them and design a fair menu.



Large: 14 in, 8 slices

Learning Goals

- I can find segment areas and use chord, secant, and tangent relationships.
- I can write circle equations in standard and general form.
- I can design a pricing model and justify it.

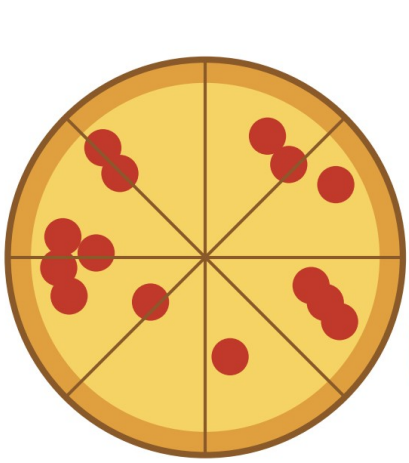
Project Timeline

When	What you will do
Day 1	Value & Shapes
Day 2	Circle Theorems
Day 3	Design a Pricing Model
Final	Pizza Menu Proposal + reflection

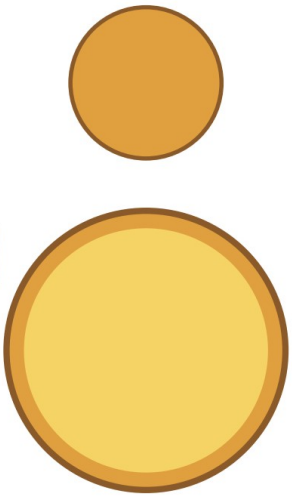
Part 1: Value & Shapes

Day 1 · Exact answers first.

1. Show that the ratio of the areas of two pizzas equals the square of the ratio of their diameters. Use the large and medium.



Large: 14 in, 8 slices





2. A slice is cut along a straight chord (not through the center) that spans a 90° arc of a large pizza. Find the area of this segment.

3. Find the diameter of a single pizza with the same area as two mediums.

Part 2: Circle Theorems

Day 2 · Justify each answer.

4. Two cuts (chords) cross inside a pizza. One is divided into 4 in and 6 in; the other has a 3-in piece. Find the other piece.

5. From a point outside the pizza, a tangent segment is 6 in long. A secant from the same point passes through 4 in outside the pizza before reaching the crust. Find the length of the secant inside the pizza.

6. The delivery zone is $x^2 + y^2 - 6x + 4y - 12 = 0$. Write it in standard form and find the center and radius.

7. Is the address (7, 2) inside the zone? What about (1, 1)?

Part 3: Design a Pricing Model

Day 3 · Write a pricing formula P(d) for a pizza of diameter d.

8. Write P(d) = a + b · (area) with a base cost a and a price per square inch b. Choose a and b and justify them.

9. Show that with a base cost, larger pizzas are a better value per square inch.

10. Present a menu with three sizes and your formula.

Final Product: Pizza Menu Proposal

Your final product must include:

- Area ratio proof
- Segment area and theorem solutions
- Delivery zone in standard form
- Pricing model and menu

Reflection

R1. Why do pizza shops usually charge a lower price per square inch for larger sizes?

R2. How do you complete the square to find a circle's center?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Prices are fair (similar price per square inch) and every circle calculation is correct. Goes beyond the minimum with a creative, realistic choice.	Prices are fair (similar price per square inch) and every circle calculation is correct.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Menu proposal shows each size's area, price per square inch, slice sizes, and a pricing recommendation. Polished, organized, and visually engaging.	Menu proposal shows each size's area, price per square inch, slice sizes, and a pricing recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.